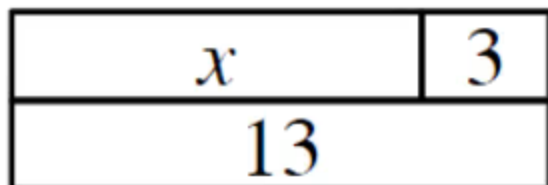




Solving simple linear equations with an additive step

- 1 What is the value of the unknown in this bar model? $x =$ _____ Fill in the blank



- 2 Which best describes the mistake that has been made when solving the equation below? Tick **1** correct answer

$$\begin{aligned}
 x - 8 &= 15 \\
 x - 8 + (-8) &= 15 + (-8) \\
 x &= 7
 \end{aligned}$$

- ☐ 15 add (-8) does not equal 7
- ☐ The same operation has not been applied to both sides of the equation.
- ☐ The correct additive inverse has not been added.
- ☐ The (-8) has not been added to the x term and the constant term on the LHS.

- 3 The solution to the equation $x - 5 = 15$ is when $x =$ _____ Fill in the blank

- 4 Which of these shows the correct working to solve the equation $x + 3.8 = 2.3$ in one additive step? Tick **1** correct answer

$$\begin{aligned}
 x + 3.8 &= 2.3 \\
 x + 3.8 + (-3.8) &= 2.3 \\
 x &= 2.3
 \end{aligned}$$

$$\begin{aligned}
 x + 3.8 &= 2.3 \\
 x + 3.8 + (-3.8) &= 2.3 + (-2.3) \\
 x &= 0
 \end{aligned}$$

$$\begin{aligned}
 x + 3.8 &= 2.3 \\
 x + 3.8 + (-3.8) &= 2.3 + (-3.8) \\
 x &= -1.5
 \end{aligned}$$

$$\begin{aligned}
 x + 3.8 &= 2.3 \\
 x + 3.8 + (-2.3) &= 2.3 + (-2.3) \\
 x - 1.5 &= 0
 \end{aligned}$$

☐☐☐☐

5 The solution to the equation $x + 7.2 = 14.6$ is when $x =$ _____ Fill in the blank

6 Which of these is the solution to the equation $x - \frac{2}{3} = \frac{5}{6}$? Tick **1** correct answer

☐ $x = \frac{1}{6}$

☐ $x = \frac{3}{4}$

☐ $x = \frac{7}{9}$

☐ $x = 1$

☐ $x = \frac{3}{2}$

